Open problems:
Quantum pure states are not enough

Open problems raised by the technical brief in this directory, and by its Lean formalization (see `formalization-log.tex`). Items marked [BLOCKS VERIFICATION] stand in the way of checking a main claim of the brief as the brief states it: a *formalization gap* can be closed in Lean alone, while a *brief issue* needs the brief to be corrected or made precise first.

Theorem 5 is verified, and Theorem 3 for $\dim \mathcal{H} \geq 2$. Example 2 is verified at the level of topology and measure and, on the Riemann sphere, of the symplectic form, with the correction of the item "Hamiltonian evolution on the classical qubit". Theorem 1 is not: see the two items marked below.

From the brief

1. [BLOCKS VERIFICATION: BRIEF ISSUE] **The range of Theorem 1.** The theorem claims consistency for $1 \leq \dim V < \infty$, while its proof uses $2 \leq \dim X$. Making it precise needs four choices.

   - *Dimension* 0. The range starts at 1 if a 0-dimensional symplectic manifold ($n = 0$ degrees of freedom) is allowed, and at 2 if not. For $\dim V = 1$ the state space is a single point, and $P(V)$ has two points exactly when $2 \leq \dim V$ (`subsingleton_projectivization`, `nontrivial_projectivization_iff`).
   - *Theorem 3 depends on it.* If a point is allowed, Theorem 3 is false for $\dim \mathcal{H} = 1$: the CST ensembles of a point and the density operators of $\mathbb{C}$ are both a single point, and the constant map identifies them (`cst_qeDens_consistent_of_finrank_one`). So the brief cannot keep both $1 \leq \dim V$ in Theorem 1 and Theorem 3 for every $\mathcal{H}$.
   - *The empty space.* For $\dim V = 0$, $P(V)$ is empty (`isEmpty_projectivization`), which is a symplectic manifold of every dimension. Either lower bound, 1 or 2, needs CST to exclude it, for instance by requiring an ensemble.
   - *What "represented by" keeps.* If X and $P(V)$ need only be in bijection, the "only if" is false: $P(\ell^2)$ and the sphere $P(\mathbb{C}^2)$ both have $\mathfrak{c}$ points (`nonempty_projectivization_equiv`), and the sphere's symplectic structure can be carried over. The theorem needs V Hausdorff and X to carry the quotient topology of $P(V)$; finiteness then comes from CST's Liouville measure ω^n with finite n, not from X being symplectic (Log, "Problems found").

   Proposed: exclude dimension 0 and the empty space, so that Theorem 1 reads $2 \leq \dim V < \infty$ for a Hausdorff V with $X = P(V)$ as topological spaces. Theorem 3 and its Lean proof then stand as they are. The alternative keeps $1 \leq \dim V$, also excludes the empty space and restricts Theorem 3 to $\dim \mathcal{H} \geq 2$. (Log, "Problems found", first three items.)

2. **Ensembles supported on a subspace.** The proof of Theorem 3 compares the ensembles with support in a two-dimensional subspace A. For $\dim \mathcal{H} \geq 3$ no CST ensemble has support in $P(A)$, which has Liouville measure zero. Whether the brief means the classical ensembles of the subsystem $P(A)$, with its own Liouville measure, is open. The formalization compares the whole ensemble spaces instead. (Log, "Problems found", second item.)

3. **Hamiltonian evolution on the classical qubit.** Example 2 identifies Hamiltonian evolution with the diffeomorphisms that conserve the volume. Area-preserving reflections of the sphere are not Hamiltonian, so the area form should be meant. Lean checks this for complex conjugation of the Riemann sphere, the Bloch reflection $a_1 \mapsto -a_1$ (`conjSphere_preserves_area_reverses_form`, `globalFlow_ne_conjSphere`). The correct statement for the sphere, that the area-form-preserving diffeomorphisms isotopic to the identity are Hamiltonian because $H^1(S^2) = 0$, needs the flux homomorphism and is not formalized. That quantum evolutions are exactly the rotations is checked (`exists_unitaryAction_eq_iff`). (Log, "Problems found", Example 2 item.)

From the formalization

1. [BLOCKS VERIFICATION: FORMALIZATION GAP] **Theorem 1.** Neither direction of Theorem 1 is formalized: that $P(V)$ with the Fubini–Study form is symplectic for $\dim V < \infty$, and that a symplectic $P(V)$ forces $\dim V < \infty$. TauCeti has symplectic manifolds, the topology of projective space and the Fubini–Study form on the Riemann sphere $\cong P(\mathbb{C}^2)$; the Fubini–Study form on $\mathbb{C}P^n$ for $n \geq 2$ and the Liouville measure of a symplectic form are still missing upstream. Theorem 3 for $\dim \mathcal{H} = \infty$ rests on the second direction, so it is formalized only for $\dim \mathcal{H} < \infty$. The second direction needs the topology of $P(V)$; as sets it is false (`nonempty_projectivization_equiv`). (Log, "Results".)

2. **Example 2.** $P(\mathbb{C}^2)$ is homeomorphic to the sphere (`nonempty_homeomorph_sphere`), its area is $dz\, d\theta$ (`toSphere_eq_dz_dtheta`), the quantum evolutions are the rotations (`exists_unitaryAction_eq_iff`) and every Hamiltonian makes the Bloch vector precess (`exists_blochVector_precession`). Still missing: the area as the Liouville measure of the Fubini–Study form, and $\omega = dz \wedge d\theta$ in the chart, which need a Liouville measure upstream; the diffeomorphism $\mathbb{C}P^1 \cong S^2$ that compares the Fubini–Study form with the round area form; the integrated rotation of the precession; and the classical side of linear Hamiltonians, $X_{n \cdot x} = n \times x$, which needs moment maps. (Log, "Differences".)

3. **Extreme points.** The proof of Theorem 5 starts from the fact that the pure states are the extreme points of $D(\mathcal{H})$. The formalization does not need it and does not prove it. (Log, "Differences", last item.)